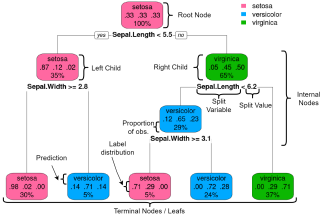
Introduction to Machine Learning

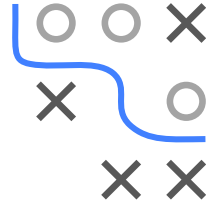
Advanced Risk Minimization
Loss functions and tree splitting



Learning goals

- Tree splitting loss vs impurity:
- Bernoulli loss $\sim$ entropy splitting
- Brier score $\sim$ gini splitting

RISK MINIMIZATION AND IMPURITY



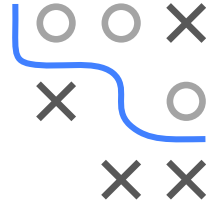
- Tree fitting: Find best way to split parent node $\mathcal{N}_0$ into child nodes $\mathcal{N}_1$ and $\mathcal{N}_2$, such that $\mathcal{N}_1 \cup \mathcal{N}_2 = \mathcal{N}_0$ and $\mathcal{N}_1 \cap \mathcal{N}_2 = \emptyset$
- Two options for evaluating how good a split is: Per node $\mathcal{N}$ compute the following:
  1. Compute impurity $\text{Imp}(\mathcal{N})$ directly from observations in $\mathcal{N}$
  2. Fit optimal constant using loss function, sum up losses for $\mathcal{N}$
- Summarize on split level:
  1. Weighted average ($n_0 = n_1 + n_2$ are number of obs in nodes)

$$\text{Imp(split)} = \frac{n_1}{n_0}\text{Imp}(\mathcal{N}_1) + \frac{n_2}{n_0}\text{Imp}(\mathcal{N}_2)$$

  2. Sum of individual losses

$$\mathcal{R}(\text{split}) = \mathcal{R}(\mathcal{N}_1) + \mathcal{R}(\mathcal{N}_2)$$

BERNOULLI LOSS MIN = ENTROPY SPLITTING



Claim: Using entropy in (1) is equivalent to using Bernoulli loss in (2)
Proof: $\mathcal{N} \subseteq \mathcal{D}$ denotes subset of observations in that node.
Risk $\mathcal{R}(\mathcal{N})$ of node $\mathcal{N}$ w.r.t. (multiclass) Bernoulli loss

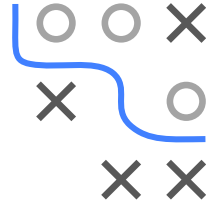
$$L\left(y, \pi(\mathbf{x})\right) = -\sum_{k=1}^{g} [y = k] \log\left(\pi_k(\mathbf{x})\right)$$

$\Rightarrow$ Optimal constant per node $\pi_k^{(\mathcal{N})} = \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} [y = k]$.

Entropy of node $\mathcal{N}$:

$$\text{Imp}(\mathcal{N}) = -\sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})}$$

BRIER SCORE MINIMIZATION = GINI SPLITTING



Claim: Using Gini in (1) is equivalent to using Brier score in (2)
Proof:
Risk $\mathcal{R}(\mathcal{N})$ of node $\mathcal{N}$ w.r.t. (multiclass) Brier score

$$L\left(y, \pi(\mathbf{x})\right) = \sum_{k=1}^{g} \left([y = k] - \pi_k(\mathbf{x})\right)^2$$

$\Rightarrow$ Optimal constant per node: $\pi_k^{(\mathcal{N})} = \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} [y = k] = \frac{n_{\mathcal{N},k}}{n_{\mathcal{N}}}$

($n_{\mathcal{N},k}$ is the number of class k observations in node $\mathcal{N}$)

Gini index of node $\mathcal{N}$:

$$\text{Imp}(\mathcal{N}) = \sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \left(1 - \pi_k^{(\mathcal{N})}\right)$$